

Answer the problems on **separate** paper. You do not need to rewrite the problem statements on your answer sheets. Do your own work. Show **all relevant steps** which lead to your solutions. Attach this question sheet to the front of your answer sheets.

1. (18 pts) For each of the following geometric series, determine:

- i. Does the series converge?
- ii. If the series is convergent, then find its sum.

a. 
$$\sum_{n=2}^{\infty} \frac{1}{18} \frac{3^{n-2}}{2^{n-1}} = \frac{1}{36} + \frac{1}{24} + \frac{1}{16} + \dots$$
      b. 
$$\sum_{n=2}^{\infty} 12 \frac{2^{n-1}}{3^n} = \frac{8}{3} + \frac{16}{9} + \frac{32}{27} + \dots$$

2. (30 pts) Determine whether the following series converge or diverge. Clearly identify which method you are employing, how you are applying your method and what your conclusions are.

a. 
$$\sum_{n=1}^{\infty} \frac{1}{n^3 + 12n}$$
      b. 
$$\sum_{n=1}^{\infty} \frac{n!}{n^2 8^n}$$
      c. 
$$\sum_{n=1}^{\infty} 4(n+1) \left(\frac{3}{4}\right)^n$$

3. (12 pts) Find the first 3 non-zero terms of the MacLaurin series for  $f(x) = \sqrt{4-x}$ .

4. (24 pts) Find the first 4 non-zero terms of the MacLaurin series for:

a. 
$$f(x) = \frac{\sin(x) + x \cos(x)}{2x}$$
      b. 
$$f(x) = (1 - e^x)(1 + e^x)$$

5. (18 pts) Using a MacLaurin series expansion for  $\sin x$ ,

- a1. approximate the value of  $\sin 0.4$  by using the first 3 terms of the expansion
- a2. approximate the value of  $\sin 0.9$  by using the first 3 terms of the expansion
- b1. find the maximum error which ensues by approximating the value of  $\sin 0.4$  by the value obtained in a1.
- b2. find the maximum error which ensues by approximating the value of  $\sin 0.9$  by the value obtained in a2.
- c1. from the maximum error in b1., determine the accuracy of the approximation in a1., i.e., determine how many decimal places are correct
- c2. from the maximum error in b2., determine the accuracy of the approximation in a2., i.e., determine how many decimal places are correct